

B1 Components of a Vector

Where bearings are given, they are in degrees East of North (so North is 000° , East is 090° , South is 180° and West 270°). For the purposes of this exercise, assume that the Earth is flat.

- B1.1 Erica the Explorer sails 250 km on a bearing of 025° . How far North is she compared to her original position?
- B1.2 A caravan has a weight of 15 000 N and sits on a ramp inclined at 6.0° to the horizontal. How big is the component of the weight which is trying to pull the caravan down the slope?
- B1.3 A laser is set up on a trolley to point west. It makes a bright spot on a wall which runs north-south. The trolley moves at 4.5 m s^{-1} on a bearing of 320° . How fast does the the spot move along the wall?

CHAPTER B. MECHANICS

Answers

B1.1 $250 \text{ km} \times \sin 25^\circ = 226.6 \text{ km}$

B1.2 $15\,000 \text{ N} \times \sin 6^\circ = 1568 \text{ N}$

B1.3 $4.5 \text{ m s}^{-1} \times \cos 320^\circ = 3.447 \text{ m s}^{-1}$